Name:

Course ID-Section: IT228-G__

Laboratory Activity 5

Laboratory Activity 5A

Using Traceroute to Discover the Network

b. What address did the command return for **b2server**? 128.107.64.254

e. What is different about the **traceroute** command on the router compared to **tracert** on the PC?
one less hop since the command is originating from R4, the order of the fields in the output is different listing the IP address in the first column.

What is the significance of R4 to Sales? It is the default gateway for Sales

f. Based on the output of the command, which interface is used to reach the next device in the list output from the **tracert** command?Interface S0/0/0 is connected to the 64.100.150.0 network.

g. What is the name of the device to which you are connected? Tier3a

i. Based on the **show ip route** command output, which interface is the exit interface for the next IP address listed in your original **tracert** output? GigabitEthernet0/0

j. What is the hostname of the current device? ISP-Tier3b

What networks are connected directly to this router? Answer here

Which interfaces connect the devices between trace route 2 and trace route 3? Answer here

k. What is the name of the device? Answer here

Addressing Scheme Table Documentation

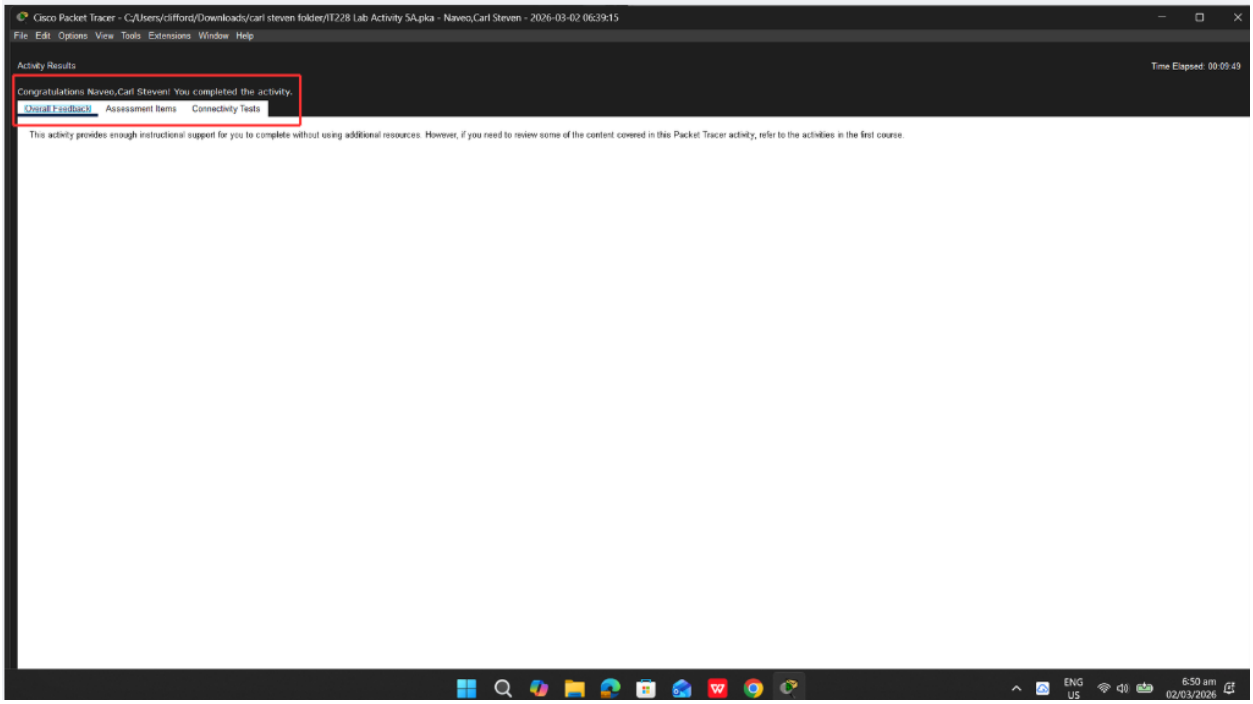
Trace Route ID	Device	Interface	Address	Subnet Mask
-	Sales	NIC	172.16.0.x (DHCP)	255.255.255.0
1	R4	G0/0	172.16.0.0	255.255.255.0
		S0/0/0	64.100.150.1	255.255.255.252
		S0/0/1.1	64.100.200.1	255.255.255.252
2	ISP-Tier3a	G/0	64.104.22.1	255.255.255.252
		G0/1	64.104.223.1	255.255.255.252
		S0/0/0	64.100.100.2	255.255.255.252
		S0/1/0	64.100.150.2	255.255.255.252

3	ISP-Tier3b	G0/1	64.104.222.2	255.255.255.252
		G0/2	64.100.8.1	255.255.255.0
		F0/1	128.107.46.1	255.255.255.0
		F/0/2	64.104.222.5	255.255.255.252
4	B2-R1	G0/0	64.104..222.6	255.255.255.252
		G0/1	128.107.64.1	255.255.255.0
5	b2server.pt.pka	NIC	128.107.64.254	255.255.255.0

Topology Documentation

Attach a **hand-drawn** topology of the new branch location. Include your name, course code and section, as well as the date of completion.

Screenshot of Check Results (with the congratulatory message that includes your name on a red box):



Laboratory Activity 5B

Configuring and Verifying a Small Network

<u>Part 2: Gather Information with Show Commands</u>

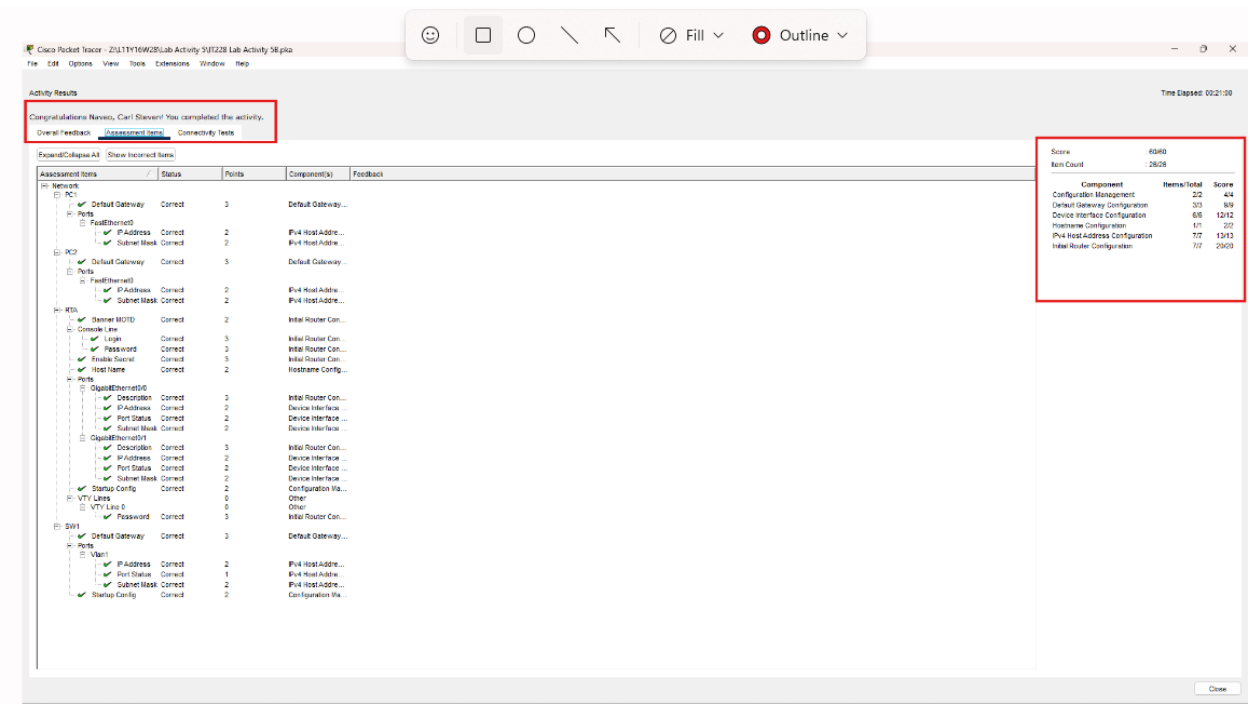
Step 1:

a. Which commands display the status of the port? Show ip interfaces brief, show interfaces, show ip interfaces
b. Which command shows only the IP address (no subnet mask or prefix)? Show ip interface brief
c. Which command displays the description configured on the interface? Show interfaces
d. Which command displays the IP broadcast address? Show interfaces
e. Which command displays the MAC address of the interface? Show interfaces

Step 2:

a. How many networks are known by the router based on the output of the **show ip route** command? 2 – 10.10.10.0/24 & 10.10.20.0/24
b. What does the **L** at the beginning of the lines within the routing table represent? Local Connection
c. What does the /32 prefix listed in the route table indicate? The Host address of the interface

Step 3:

a. On **RTA**, shut down the Gigabit Ethernet 0/0 interface and issue the **show ip route** command. How many networks are displayed in the routing table now? 1 – 0.10.20.0/24
b. Attempt to ping PC1. Was the ping successful? No
c. Issue the **show ip interface brief** command. What is the status of the Gigabit Ethernet 0/0 interface? Administratively Down
d. Reactivate the Gigabit Ethernet 0/0 interface. Issue the **show ip route** command. Did the routing table repopulate? Yes
What can be inferred about the interface status of routes that appear in the routing table? Interfaces must be active in order to be listed in the routing address

Screenshot of Assessment Items (with name and score on a red box):